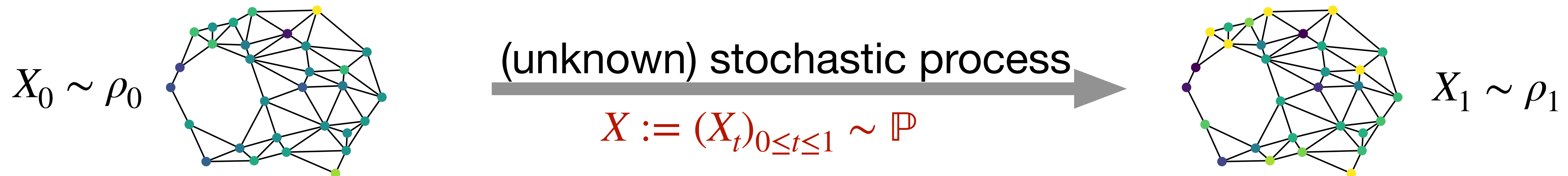


Topological signal distribution matching

- In a topological domain, e.g., a graph, signals on the node set.
- Given (empirical) signal distributions at $t = 0$ and at $t = 1$



- Assume some prior (reference) process $Y \sim \mathbb{Q}_{\mathcal{T}}$ — — **topology-aware**

Topological Schrödinger Bridge Problem

$$\min D_{KL}(\mathbb{P} || \mathbb{Q}_{\mathcal{T}}) \text{ s.t. } \mathbb{P}_0 = \rho_0, \mathbb{P}_1 = \rho_1$$

Gaussian special case:

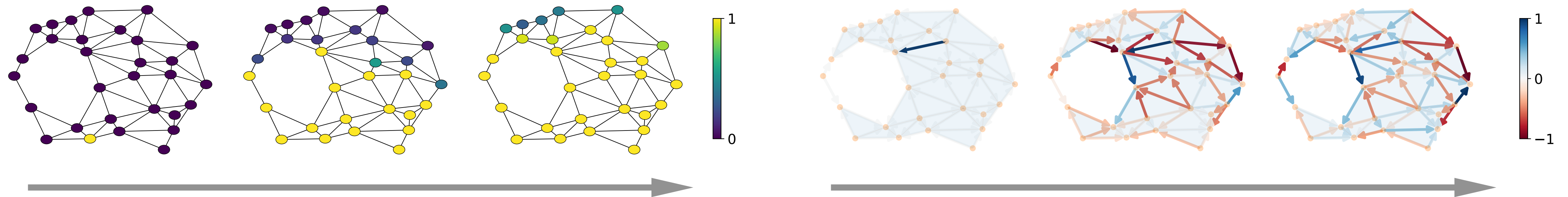
$$\rho_0 \sim N(\mu_0, \Sigma_0(L))$$

$$\rho_1 \sim N(\mu_1, \Sigma_1(L))$$

[Yang et al. 2024 aistats]

Reference topological dynamics

- $Y \sim \mathbb{Q}_{\mathcal{T}}$: topology-aware stochastic dynamics, tractable topology-aware transition kernel 
- Topological stochastic heat diffusion: $dY_t = -cLY_t dt + g_t dW_t$



- Topological stochastic dynamics: $dY_t = f(t, Y_t; L)dt + g_t dW_t$
 - $f_t = H_t(L)Y_t + \alpha_t$ with $H_t(L)$ a topological convolution wrt Laplacian L